

Supplementary Information

Designing Any Imaging System from Natural Language:
Agent-Constrained Composition over a Finite Primitive Basis

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1 S1. Foundational Theorems from the Companion Paper

The companion paper [1] establishes two foundational results that this paper builds upon. To make the present work self-contained, we reproduce the full theorem statements, proof sketches, and key lemmas below.

1.1 S1.1. Finite Primitive Basis (FPB) Theorem

Theorem 1 (Finite Primitive Basis). *For every imaging forward model H in the designable class $\mathcal{C}_{\text{tier}}$, there exists a well-formed typed DAG $G = (V, E, \tau)$ whose node types are drawn from the canonical primitive library $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R, \Lambda\}$ (11 elements) such that G is an ε -approximate representation of H with $\varepsilon = 0.01$.*

Proof sketch. The proof is constructive. Given any forward model $H = H_K \circ H_{K-1} \circ \dots \circ H_1$, we decompose it into a sequence of physics stages and show each stage is representable by one or more FPB primitives via six *Primitive Realization Lemmas*:

Lemma 1 (Propagation Realization). Free-space wave propagation H_k is represented by the Propagate primitive $P(d, \lambda)$ via the angular spectrum representation: $P = \mathcal{F}^{-1}[T_{\text{exact}} \cdot \mathcal{F}[\cdot]]$. Error from evanescent wave truncation: $\varepsilon_{\text{evan}} = e^{-2\pi d/\lambda}$, negligible for $d \gg \lambda$.

Lemma 2 (Elastic Interaction Realization). Element-wise transmission multiplication is *exactly* represented by the Modulate primitive $M(\mathbf{m})$: $\|H_k - M(\mathbf{m})\| = 0$.

Lemma 3 (Scattering Realization). Under the first Born approximation, the scattered field is a linear functional representable by the Scatter primitive R . Single-scattering representation is exact; L -th order multiple scattering error: $\varepsilon_{\text{scat}}^{(L)} \leq (\|V\|/k)^{L+1}$ (geometric convergence).

Lemma 4 (Encoding-Projection Realization). Line-integral projection (CT) is exactly the Radon transform $\Pi(\theta)$; Fourier encoding (MRI) is exactly $F(\mathbf{k})$. Both are exact: $\varepsilon = 0$.

Lemma 5 (Detection-Readout Realization). The detection chain decomposes into ≤ 5 sequential operations, each representable by a primitive. Detector PSF error: $\varepsilon_{\text{PSF}} \leq \ell_c^2/p^2$ (crosstalk length ℓ_c , pixel pitch p). Total: $\varepsilon_{\text{det}} < 0.01$ for all 39 validated modalities.

Lemma 6 (Nonlinear-Stage Realization). All pointwise physics nonlinearities belong to 5 structural categories (exponential, logarithmic, phase wrapping, polynomial, saturation), each representable by the Transform primitive $\Lambda(f, \theta)$.

Error composition bound. Let $\gamma_k = \|H_k\|$ (linear) or $\text{Lip}(H_k)$ (nonlinear). By Lipschitz composition:

$$\sup_{\|\mathbf{x}\| \leq 1} \|H(\mathbf{x}) - H_{\text{dag}}(\mathbf{x})\| \leq \sum_{k=1}^K \varepsilon_k \prod_{j=k+1}^K \gamma_j \leq K \cdot \max_k(\varepsilon_k) \cdot B^{K-1}$$

With empirical maximum $K=5$ and $B=4$: requires $\max_k(\varepsilon_k) \leq 3.9 \times 10^{-5}$, satisfied by all 6 lemmas. Per-modality tightening exploits the fact that most $\varepsilon_k = 0$ (elastic, encoding stages) and most primitives are norm-preserving.

Necessity. All 11 primitives are necessary: for each primitive B_i , there exists a witness modality whose representation error exceeds 0.01 when B_i is removed. Witnesses: $P \leftarrow$ ptychography, $M \leftarrow$ CASSI, $\Pi \leftarrow$ CT, $F \leftarrow$ MRI, $C \leftarrow$ lensless, $\Sigma \leftarrow$ SPC, $D \leftarrow$ all, $S \leftarrow$ MRI, $W \leftarrow$ CASSI, $R \leftarrow$ Compton ($\varepsilon = 0.34$ without), $\Lambda \leftarrow$ polychromatic CT.

Complexity bound. Node count: $|V| \leq 20$ (empirical max: 5). Depth: $\text{depth}(G) \leq 10$ (empirical max: 5).

1.2 S1.2. Triad Decomposition Theorem

Theorem 2 (Triad Decomposition). *Let $H \in \mathcal{C}_{\text{tier}}$ with measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$ and reconstruction using nominal operator H_{nom} . The reconstruction MSE decomposes as:*

$$\text{MSE} \leq \text{MSE}^{(G_1)} + \text{MSE}^{(G_2)} + \text{MSE}^{(G_3)}$$

where $\text{MSE}^{(G_1)}$ is the null-space loss (information-theoretic floor), $\text{MSE}^{(G_2)}$ is the noise-floor term (carrier budget), and $\text{MSE}^{(G_3)}$ is the mismatch-induced term (model fidelity). This decomposition is diagnostically complete: every reconstruction failure is attributable to at least one gate.

Gate 1 (Recoverability / Compression Bound). For any estimator $\hat{\mathbf{x}}(\mathbf{y})$, the minimum achievable MSE satisfies:

$$\text{MSE}_{\min}^{(G_1)} \geq \frac{1}{n} \sum_{i=1}^{n-m} \sigma_i^2(\mathbf{x})$$

where $\sigma_i^2(\mathbf{x})$ is the variance of $\mathbf{x}$ along the i -th null-space direction of H , m is the number of measurements, and n is the signal dimension. The corresponding PSNR ceiling is:

$$\text{PSNR}_{\max}^{(G_1)} = 10 \log_{10} \left(\frac{\|\mathbf{x}\|_{\infty}^2}{\text{MSE}_{\min}^{(G_1)}} \right)$$

No algorithm can exceed this ceiling without additional priors.

Gate 2 (Carrier Budget / Noise Bound). Under matched forward model and Gaussian noise $\mathbf{n} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I})$:

$$\text{PSNR}_{\max}^{(G_2)} = 10 \log_{10}(\text{SNR}) + C_{\mathcal{M}}, \quad C_{\mathcal{M}} = 10 \log_{10} \left(\frac{n \|\mathbf{x}\|_{\infty}^2}{\|\mathbf{x}\|^2} \cdot \kappa^{-2}(H) \right)$$

where $\kappa(H)$ is the condition number of H restricted to its column space. No algorithm can exceed this ceiling regardless of computational budget.

Gate 3 (Operator Mismatch / Calibration Sensitivity). For small perturbation $\delta\theta$ around true parameters θ^* :

$$\Delta \text{PSNR} \approx - \frac{10}{\ln 10} \frac{\delta\theta^{\top} \mathbf{J}^{\top} \mathbf{J} \delta\theta}{\text{MSE}_0}$$

where $\mathbf{J} = \partial(H\theta\mathbf{x})/\partial\theta|_{\theta^*}$ is the parameter Jacobian and MSE_0 is the noise-only MSE at θ^* .

Gate 3 dominance condition. For instruments operating above Gates 1 and 2 floors, Gate 3 becomes the binding constraint when $\|\delta\theta\|_{\mathbf{J}^{\top} \mathbf{J}} > (\sigma_n / \|\mathbf{x}\|_{\infty}) \cdot \kappa(H)$. Empirically, Gate 3 dominates in all 12 validated modalities across 5 carrier families: CASSI shows 13.98 dB degradation from sub-pixel mask shift; ptychography shows 42.06 dB sensitivity to probe mismatch.

Recovery ratio. $\rho = (\text{PSNR}_{\text{achieved}} - \text{PSNR}_{\text{mismatch}}) / (\text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{mismatch}})$, bounded in $[0, 1]$ under convex reconstruction loss. Across validated modalities: $\rho \in [0.4, 0.9]$, mean 0.85.

2 S2. Novel System Design Specifications

The 10 novel system designs compose FPB primitives into chains not found in prior literature. Each is specified via the 8-field `spec.md` format. The notation Φ_z denotes depth-parameterised convolution: the C primitive applied with a depth-dependent point spread function $\Phi(z)$, generated by wave propagation through a random phase diffuser with defocus modulation.

2.1 S2.1. 2D Systems

Lensless imaging ($C \rightarrow D$, compression 1:1). A thin random phase mask creates a caustic PSF with broad spatial-frequency support. Reconstruction inverts the convolution via ADMM+TV. PSNR: 43.7 dB, SSIM: 0.984.

```
modality: lensless_imaging
carrier: photon
geometry: single_shot, 128x128
object: 128x128 2D image
forward_model: Convolve(C, psf=phase_mask) -> Detect(D)
noise: Poisson I_0=5000 + Gaussian sigma=3
target: PSNR >= 40 dB
system_elements:
  source: broadband LED
  optics: random phase mask (feature_scale=2.5)
  detector: CMOS 128x128
```

SIM ($M \rightarrow C \rightarrow D$, super-resolution). Structured illumination modulates the sample with sinusoidal patterns, shifting high-frequency content into the passband. Three pattern orientations at three phases yield 9 measurements for $2\times$ resolution enhancement. PSNR: 27.7 dB.

2.2 S2.2. 3D Systems (8:1 compression)

3D Lensless ($\Phi_z \rightarrow \Sigma \rightarrow D$). A random phase diffuser creates depth-dependent PSFs through defocus-modulated wave propagation: $\text{PSF}(z) = |\mathcal{F}\{\exp(i\phi_{\text{diffuser}} + i \cdot \text{defocus}(z) \cdot r^2)\}|^2$. Each depth plane produces a unique speckle pattern; the 2D sensor measurement is the sum of all convolved depth planes. This is the C primitive instantiated per depth plane. PSNR: 20.3 dB at 8:1 compression.

```
modality: 3d_lensless
carrier: photon
geometry: single_shot, 128x128, n_depths=8
object: 128x128x8 3D volume (sparse depth planes)
forward_model: Convolve_z(C, psf=diffuser(z)) -> Sum(Sigma) -> Detect(D)
noise: Poisson I_0=5000 + Gaussian sigma=3
target: PSNR >= 15 dB
system_elements:
  source: broadband LED
  optics: random phase diffuser (feature_scale=1.0, defocus_max=40)
  detector: CMOS 128x128
```

Temporal-coded lensless ($M \rightarrow C \rightarrow \Sigma \rightarrow D$). A binary DMD mask modulates each of 8 video frames before diffuser convolution and temporal integration during a single exposure. The mask provides per-pixel temporal diversity. PSNR: 31.6 dB.

Spectral lensless ($M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$). A coded mask followed by prism dispersion separates 8 spectral bands before diffuser convolution and spectral integration. PSNR: 36.3 dB.

2.3 S2.3. 4D Systems (16:1 compression)

4D Spectral-Depth ($M \rightarrow W_\lambda \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). Combines coded mask, spectral dispersion, and diffuser depth encoding to recover a (z, λ) datacube ($4 \times 4 = 16$ planes) from a single 2D shot. PSNR: 23.9 dB.

4D Temporal DMD ($M \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). DMD binary masks provide temporal coding; diffuser PSFs encode depth. Recovers (z, t) datacube at 16:1. PSNR: 25.4 dB.

4D Temporal Streak ($M \rightarrow W_t \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). Passive temporal dispersion (streak camera) replaces active DMD coding. A fixed mask provides spatial diversity. PSNR: 25.3 dB.

2.4 S2.4. 5D Systems (64:1 compression)

5D Full DMD ($M \rightarrow W_\lambda \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). Full 5D recovery of (x, y, z, λ, t) from a single shot at $4 \times 4 \times 4 = 64:1$ compression using coded mask + spectral dispersion + diffuser depth encoding. PSNR: 29.9 dB.

5D Full Streak ($M \rightarrow W_\lambda \rightarrow W_t \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). Replaces DMD with passive streak temporal dispersion. The longest chain in the FPB framework (5 operators before detection). PSNR: 29.9 dB.

2.5 S2.5. Design Patterns

Three patterns emerge from the novel designs:

1. **Active modulation (DMD) > passive dispersion:** Binary masks provide per-pixel measurement diversity that continuous dispersion cannot match. DMD-based temporal coding (31.6 dB) outperforms streak-based (25.3 dB) at equal compression.
2. **Algorithm choice is critical for compressive systems:** FISTA+TV or R-L dominate across novel modalities. Inter-method PSNR CoV reaches 42–48% for compressive systems vs. <6% for well-conditioned ones.
3. **Graceful compression degradation:** 1:1 $\rightarrow$ 43.7 dB; 8:1 $\rightarrow$ 20–36 dB; 16:1 $\rightarrow$ 24–25 dB; 64:1 $\rightarrow$ 29–30 dB.

3 S3. Maskless Control Group

To validate that spatial modulation (M) is a prerequisite for compressive dispersion-based imaging, we tested 9 configurations that combine depth encoding (Φ_z) with spectral (W_λ) and/or temporal (W_t) dispersion but *without* a coded mask. All 9 configurations compile through the structural compiler (Gates 1–5) but are flagged by the Judge Agent (Gate 1, recoverability) as severely ill-conditioned:

- Condition number $> 10^6$ for all maskless configurations
- Reconstruction PSNR: 6–9 dB (near noise floor)

- Adding a binary coded mask (M) improves PSNR by 5–15 dB at equal compression

This confirms a fundamental principle: dispersion operators (W) provide only global spectral/temporal diversity, while coded masks (M) provide per-pixel spatial diversity. Both are necessary for well-conditioned compressive recovery.

4 S4. Encoder Comparison for Depth Encoding

Three physically realisable depth-encoding mechanisms were compared across 5 modulation combinations ($3 \times 5 = 15$ configurations total):

Diffuser (Φ_z^{diff}): Random phase plate with defocus-dependent PSFs. $\text{PSF}(z) = |\mathcal{F}\{\exp(i\phi_{\text{random}} + i \cdot \text{defocus}(z) \cdot r^2)\}|^2$.

Multi-lens (L_z): Microlens array producing depth-dependent parallax sub-pixel shifts.

Meta-surface (Ψ_z): Engineered phase pattern (cubic + trefoil + astigmatism aberrations) with defocus modulation.

Table 1: Depth encoder comparison: best PSNR (dB) across 3 encoders $\times$ 5 modulation combinations. $N_z=8$ depth planes, $N_\lambda=4$ spectral bands, $N_t=4$ temporal frames, image size 128×128 .

Encoder	3D only (8:1)	+Mask (8:1)	+M+W $_\lambda$ (32:1)	+M+W $_t$ (32:1)	+M+W $_\lambda$ +W $_t$ (128:1)
Diffuser	17.0	20.3	24.6	26.9	38.1
Multi-lens	16.6	44.1	41.0	36.1	38.7
Meta-surface	16.6	16.2	24.6	25.8	38.0

Key findings:

- Without coded mask: all three encoders perform similarly (16.6–17.0 dB), limited by the ill-conditioning of depth-only recovery at 8:1 compression.
- Multi-lens + mask dominates: parallax shifts are simple, well-conditioned operators (sub-pixel translations). Combined with a binary mask’s per-pixel diversity, the composite forward operator is nearly diagonal, yielding 44.1 dB.
- At maximal compression (128:1, 5D), all encoders converge to ~ 38 dB because the mask and dispersion operators dominate the measurement diversity.

PSF diversity analysis (mean inter-depth cross-correlation):

- Diffuser: 0.29 (moderate diversity, structured speckle)
- Multi-lens: 0.17 (lowest correlation, shift-based diversity)
- Meta-surface: 0.46 (highest correlation, engineered but similar patterns)

5 S5. Reconstruction Details

5.1 S5.1. Algorithms

Five reconstruction algorithms were evaluated for each novel modality:

Wiener deconvolution: Frequency-domain least-squares with noise regularisation. Fastest (<0.1 s) but lowest quality for compressive systems.

GAP-TV: Generalised alternating projection with total variation regularisation [2]. Alternates between data projection and TV denoising.

FISTA+TV: Fast iterative shrinkage-thresholding with TV proximal operator [3]. Nesterov-accelerated gradient descent with $O(1/k^2)$ convergence.

ADMM+TV: Alternating direction method of multipliers with TV splitting [4]. Variable splitting separates data fidelity from regularisation.

Richardson-Lucy (R-L): Maximum-likelihood deconvolution for Poisson noise. Multiplicative updates preserve non-negativity.

5.2 S5.2. Forward Model Implementation

All forward models are implemented as linear operators with explicit adjoint computation. The measurement for a multi-dimensional modality with depth encoding, mask, and dispersion is:

$$y = \sum_{z,\lambda,t} D \left[\text{shift}_{(\Delta x_\lambda, \Delta y_t)} (M \odot \text{conv}(x_{z,\lambda,t}, \text{PSF}(z))) \right] + n$$

where D is detection, M is the binary mask, conv denotes convolution with depth-dependent PSF, shift implements spectral/temporal dispersion as lateral shifts, and n is Poisson + Gaussian noise.

6 S6. LLM Implementation Details

6.1 S6.1. Model and Prompting

All three agents (Plan, Judge, Execute) use Claude Sonnet 4 (Anthropic, `claude-sonnet-4-20250514`) as the backbone LLM. The system prompt provides each agent with:

- The 11 FPB primitives with typed signatures and physical semantics
- The 173-modality registry (modality name, carrier, canonical chain, typical parameters)
- The `spec.md` schema with field descriptions and 3 worked examples (CT, MRI, CASSI)
- Agent-specific role instructions (Plan: generate specification; Judge: validate via Triad gates; Execute: select algorithm and predict quality)

Temperature is set to 0 for reproducibility. The Plan Agent uses chain-of-thought prompting: it first identifies the carrier family, then selects primitives left-to-right along the physical signal path, then fills in parameters from the registry or physical constraints. The Judge Agent applies deterministic gate checks (no LLM inference for Gates 1–4); only Gate 5 (cost estimation) and Gate 6 (FPB fidelity) use LLM reasoning.

6.2 S6.2. Reproducibility and Robustness

Three `spec.md` files (CT, MRI, CASSI) were independently generated 5 times each; all 15 specifications were identical, yielding PSNR within ± 0.3 dB. The 4% misspecification rate (7/173 modalities) occurs exclusively for modalities with ambiguous natural-language descriptions (e.g., “4D-STEM” could refer to scanning or convergent-beam modes).

LLM dependency is a limitation. The agents have been tested only with Claude Sonnet 4. Because 4 of 6 Judge gates are deterministic (no LLM inference), and the Execute Agent’s algorithm selection is from a fixed catalog, the LLM dependency is concentrated in the Plan

212 Agent’s specification generation. Cross-LLM robustness testing is future work; we expect that
213 any LLM capable of structured JSON output and physics reasoning would produce comparable
214 results for well-documented modalities, but novel or ambiguous modalities may show higher
215 variance.

216 References

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